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CHALLENGES TO SUSTAINABLE DEVELOPMENT

India’s Vulnerability Profile: Geographical Zones and Disaster Susceptibility

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India is one of the most disaster-prone countries on the planet. Every year, roughly 3 crore (30 million) people across the country are affected by natural calamities ranging from floods and droughts to earthquakes and cyclones. This isn’t a random occurrence—it’s a direct consequence of India’s unique geographical position, diverse topography, and climatic conditions. Understanding where and why these disasters strike is essential for building resilience, protecting lives, and ensuring that development gains are not repeatedly wiped out.

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India’s recurring natural disaster pattern

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The statistics paint a stark picture. [As per India's seismic zone map](#), over 59% of the country's landmass faces moderate to severe seismic hazard. More than 40 million hectares are flood-prone, representing about 12% of India's total geographical area. Approximately 8% of the country's area, primarily along its eastern and western coastlines, is susceptible to cyclones. And a staggering 68% of cultivable land is [vulnerable to drought](#).

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Between 2000 and 2019, India recorded [321 catastrophic events and suffered nearly \\$80 billion in economic losses](#). The country ranks third globally in the number of natural disasters, behind only China and the United States. This multi-hazard exposure creates overlapping and compounding risks that make disaster management an especially complex challenge.

Five geographical disaster zones of India

India's disaster vulnerability becomes clearer when the country is divided into five distinct geographical zones, each with its own characteristic disaster profile. These zones are shaped by topography, elevation, climate patterns, and proximity to coastlines or mountain ranges.

The northern mountain region

The Himalayan belt stretching across northern India—covering states like Jammu & Kashmir, Himachal Pradesh, Uttarakhand, Sikkim, and the northeastern hill states—is among the most hazard-prone areas in the country. This region sits at the collision boundary of the Indian and Eurasian tectonic plates, making it highly susceptible to earthquakes. In fact, the entire Himalayan belt is considered prone to great earthquakes exceeding magnitude 8.0.

In addition to seismic activity, this zone faces [frequent landslides triggered by heavy monsoon rainfall](#), particularly in the Western Ghats and northeastern hill ranges. Cold waves during winter months add another layer of hazard. The 2024 Wayanad landslides in Kerala, which resulted from extreme rainfall of over 572 mm within 48 hours, demonstrated how devastating these events can be—[over 336 fatalities were recorded](#) in that single event.

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Glacial lake outburst floods (GLOFs) are an emerging threat in this zone. As Himalayan glaciers continue to recede—having shrunk by about 21% since 1962—unstable glacial lakes form and can breach suddenly, causing catastrophic downstream flooding.



Flooding is India’s most common natural disaster. [On an annual average, 7.5 million hectares are flooded and approximately 1,600 lives are lost.](#) The heavy southwest monsoon rains cause rivers like the Brahmaputra to overflow, inundating surrounding areas. While these floods historically provided natural irrigation and soil fertilisation for agriculture, they now increasingly cause massive displacement and economic loss due to growing populations settled in floodplains.

Flash flooding has become more common in central India over recent decades, coinciding with rising temperatures and changing precipitation patterns. The 2024 monsoon season was a grim reminder-floods in Bihar alone displaced over 36,000 people and affected 1.2 million across 12 districts.

The Deccan plateau

The Deccan plateau, covering much of central and southern peninsular India, is primarily a **drought-prone zone**. States like Maharashtra, Karnataka, Andhra Pradesh, and Telangana regularly experience water stress, particularly in rain-shadow areas where monsoon rainfall is unreliable.

Drought may occur less frequently than floods, but its impact is far more widespread. [A single drought event can impact an average of 75 million people](#)—more than floods and cyclones combined in terms of affected populations. Drought conditions severely disrupt agriculture, threaten food security, and push rural communities into deeper poverty.

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While the Deccan was historically considered relatively safe from seismic activity, events like the 1993 Latur earthquake forced a reassessment. Parts of peninsular India have since been reclassified into higher seismic zones, adding earthquake risk to the region’s drought vulnerability.

The western desert region

The Thar Desert, spanning Rajasthan and parts of Gujarat, is another **severely drought-susceptible zone**. Rainfall here is scanty and erratic, and water scarcity is a chronic challenge. This region faces some of the most extreme temperature variations in India, with scorching summers and cold desert nights.

The 2024 summer saw India record some of its [hottest-ever heatwaves, with temperatures exceeding 49°C in northern regions](#). While the western desert has always endured extreme heat, climate change is intensifying these events and making them more frequent. Drought here is aggravated by the El Niño-Southern Oscillation (ENSO) phenomenon, which disrupts monsoon patterns and further reduces already scarce rainfall.

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The coastal zone

India’s extensive coastline—stretching approximately [7,516 km across 13 states and union territories](#)—creates a distinct and critical disaster zone. About 5,700 km of this coastline is continuously exposed to tropical cyclones, making coastal communities highly vulnerable.

The Bay of Bengal generates roughly 80% of the cyclones that impact India, with the eastern coast states of Odisha, Andhra Pradesh, West Bengal, and Tamil Nadu bearing the brunt. The ratio of cyclones in the Bay of Bengal to the Arabian Sea is approximately 4:1. Sea erosion, tidal surges, and tsunami risk further compound coastal vulnerability—as the devastating 2004 Indian Ocean tsunami tragically demonstrated.

Coastal vulnerability is increasing as [roughly 71 lakh dwellings are already located within 100 metres of the Indian coastline](#). Sea levels along India’s coast have risen by 8.5 cm over the past 50 years, and projections suggest 36 million Indians could be living in areas experiencing chronic flooding by 2100.

Historical disaster impact and key events

India’s disaster history is long and devastating. The country’s geo-climatic conditions, combined with its vast population and long coastline, have made it traditionally vulnerable to catastrophic events that set back development progress by years or even decades.

Major flood years—1955, 1971, 1977, 1978, 1988, 1998, 2004—have each caused enormous economic and human losses. [During the last 15 years alone, India experienced 10 major earthquakes resulting in over 20,000 deaths](#). The 2001 Gujarat earthquake, the 2004 tsunami, the 2013 Uttarakhand flash floods, and the 2018 Kerala floods are just some of the disasters that have left deep scars on affected regions.

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More recently, the first half of 2024 exposed how unprepared India remains for escalating climate risks. Record-breaking heatwaves, devastating floods across Assam and Manipur triggered by Cyclone Remal, and the catastrophic Wayanad landslides all occurred within months of each other. As a [Toda Peace Institute analysis noted](#), these events suggest climate change impacts are no longer “impending”—they are already altering lives across the Indian subcontinent.

Who suffers most: the inequality dimension of disasters

Natural disasters do not affect all people equally. In India, the poor and under-privileged are disproportionately impacted due to a combination of locational disadvantage and economic defenselessness.

Marginalized communities—including Dalits, tribal populations, women-headed households, and people with disabilities—often inhabit the most hazard-exposed locations: floodplains, unstable hillsides, or eroding coastlines. They settle in these areas not

Historical settlement patterns and economic marginalisation push vulnerable groups into high-risk areas. Slums and informal settlements in urban areas are often built on floodplains or unstable slopes. Rural communities may farm on riverbanks or depend on rain-fed agriculture in drought-prone zones. This geographic exposure means the same flood or cyclone causes far greater damage to poor communities than to wealthier ones.

Economic defenselessness

Without savings, insurance, or social safety nets, economically vulnerable populations cannot absorb disaster losses or finance recovery. [A study published in the Journal of Safety Science and Resilience](#) found that West Bengal and Odisha are among the most vulnerable coastal states, with marginalized groups living in kutchha (mud or bamboo) houses unable to withstand cyclonic winds. Each successive disaster pushes these communities further into poverty, creating a vicious cycle of vulnerability.

Limited resilience capacity

The ability to bounce back to pre-disaster normality depends heavily on resources, knowledge, and institutional support—all of which are unevenly distributed. Early warning systems and disaster preparedness measures are often less effective in remote or marginalised regions. The result is that disasters with similar physical characteristics produce vastly different human outcomes depending on the socioeconomic profile of the affected community.

Climate change as a threat multiplier

Climate change is making India's disaster vulnerability worse across all five geographical zones. Glacial melting in the Himalayas is increasing flood and GLOF risks. Monsoon patterns are becoming more erratic, with longer dry spells punctuated by intense rainfall events. Cyclones are gaining intensity. Heatwaves are lasting longer and reaching higher temperatures. Sea levels are rising along vulnerable coastlines.

India's [27 out of 29 states and 7 union territories are susceptible to natural hazards](#). With scientific evidence indicating that extreme weather events will become more frequent and intense in the Indian subcontinent, building adaptive capacity is no longer optional—it is urgent.

Effective disaster risk reduction must account for local geographical, climatic, and socioeconomic conditions. Zone-specific strategies—flood management in the Indo-Gangetic plains, drought resilience in the Deccan and western desert, earthquake-resistant construction in the Himalayas, and coastal protection along vulnerable shorelines—need to work alongside broader efforts to reduce socioeconomic inequality that amplifies disaster impact.

What do you think? Given that India's poorest communities bear the heaviest burden of disasters, should disaster management policy prioritise socioeconomic resilience—building alongside physical infrastructure? And how can zone-specific approaches be better integrated into India's broader sustainable development planning?



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